DECEMBER 2024

Data Quality, Management and Governance

Short course

SCIENTIA ET PRATIQUE

MTF

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Data Quality, Management and Governance

Unlocking the power of Data Quality, Management and Governance

Course: Data Quality, Management and Governance

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Module 1: Introduction to Data Governance

What is Data Governance?

Definition and Scope:

- Data governance involves the management and oversight of data to ensure its accuracy, quality, and security.

Evolution of Data Governance:

- Developed from basic data management practices to comprehensive frameworks ensuring data integrity and compliance.

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Module 1: Introduction to Data Governance

Importance of Data Governance

- Ensuring Data Accuracy, Privacy, and Security:
  - Protects against data breaches and ensures data is reliable.
- Supporting Business Decision-Making:
  - High-quality data is essential for informed decision-making.
- Compliance with Regulations and Standards:
  - Ensures adherence to legal and regulatory requirements.

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Module 1: Introduction to Data Governance

Key Principles of Data Governance

- Accountability:
  - Clearly defined roles and responsibilities for data management.
- Integrity:
  - Ensuring the accuracy and consistency of data.
- Transparency:
  - Clear documentation and communication of data policies and procedures.

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Module 1: Introduction to Data Governance

Main Components of Data Governance

- Data Policies:
  - Rules and guidelines for data management and usage.
- Data Standards and Procedures:
  - Established norms for data handling to ensure consistency.
- Data Stewardship:
  - Designated roles responsible for managing and protecting data assets.

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Module 1: Introduction to Data Governance

Benefits of Effective Data Governance

- Improved Data Quality:
  - Higher accuracy and reliability of data.
- Enhanced Decision-Making Capabilities:
  - Better data leads to more informed and effective decisions.
- Regulatory Compliance:
  - Adherence to laws and standards.
- Reduced Risks and Costs:
  - Minimises data breaches and associated costs.
- Increased Operational Efficiency:
  - Streamlined data processes enhance productivity.

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Module 1: Introduction to Data Governance

Overview of Data Governance Frameworks

- Introduction to Various Frameworks:
  - Overview of frameworks like DAMA-DMBOK, COBIT, and ISO 38500.
- Role in Establishing Data Governance Structure:
  - Frameworks provide a structured approach to implementing data governance.
- Benefits of Adopting a Framework:
  - Ensures comprehensive and consistent data governance practices.

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Module 1: Introduction to Data Governance

Example Frameworks: DAMA-DMBOK, COBIT, ISO 38500

- DAMA-DMBOK:
  - Key concepts and knowledge areas.
- COBIT:
  - Components and support for data governance.
- ISO 38500:
  - Principles and model for IT governance.

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Module 2: Data Governance Frameworks

Overview of Popular Frameworks

- DAMA-DMBOK:
  - Key concepts and knowledge areas
  - Application and benefits
- COBIT:
  - Overview and components
  - How COBIT supports data governance
- ISO 38500:
  - Governance of IT for the organisation
  - Principles and model

Module 2: Data Governance Frameworks

Comparison of Frameworks

- Strengths and Weaknesses:
  - DAMA-DMBOK: Comprehensive but complex
  - COBIT: Strong IT alignment but can be rigid
  - ISO 38500: High-level principles but lacks detailed guidance
- Key Differences and Similarities:
  - Focus areas, depth, and application scope
- Best Use Cases:
  - DAMA-DMBOK: Data-centric organisations
  - COBIT: IT-intensive enterprises
  - ISO 38500: Organisations seeking IT governance alignment

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Module 2: Data Governance Frameworks

Selecting the Right Framework for Your Organisation

- Assessing Organisational Needs:
  - Identify data governance requirements and goals
- Evaluating Framework Compatibility:
  - Match framework strengths to organisational needs
- Making an Informed Decision:
  - Consider scalability, flexibility, and resource availability

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Module 2: Data Governance Frameworks

Implementation Strategies

- Planning and Preparation:
  - Define objectives and scope
  - Secure stakeholder buy-in
- Step-by-Step Implementation:
  - Develop a detailed implementation plan
  - Assign roles and responsibilities
  - Conduct training and awareness programs
- Monitoring and Continuous Improvement:
  - Establish metrics and KPIs
  - Regularly review and update governance practices
  - Foster a culture of continuous improvement

Module 3: Data Governance Roles and Responsibilities

Key Roles in Data Governance

- Data Owners:
  - Overall data management
  - Ensuring data quality and compliance
- Data Stewards:
  - Responsibilities in managing data assets
  - Ensuring adherence to data policies
- Data Custodians:
  - Technical management of data
  - Safeguarding data assets

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Module 3: Data Governance Roles and Responsibilities

Responsibilities and Skill-sets Required

- Defining Roles and Responsibilities:
  - Clear role definitions
  - Aligning responsibilities with organisational goals
- Essential Skills for Data Governance Roles:
  - Data management and quality assurance
  - Regulatory knowledge and compliance
  - Communication and collaboration
- Training and Development:
  - Continuous learning opportunities
  - Certification and professional development programs

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Module 3: Data Governance Roles and Responsibilities

Building an Effective Data Governance Team

- Team Structure and Composition:
  - Hierarchical vs. matrix structures
  - Roles and interactions within the team
- Fostering Collaboration and Communication:
  - Cross-functional collaboration
  - Regular meetings and updates
- Encouraging a Culture of Data Governance:
  - Promoting data governance awareness
  - Recognising and rewarding good practices

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Module 3: Data Governance Roles and Responsibilities

Role of Chief Data Officer (CDO)

- Importance of the CDO in Data Governance:
  - Strategic leadership and vision
  - Driving data governance initiatives
- Key Responsibilities and Impact:
  - Overseeing data management practices
  - Ensuring data quality and compliance
  - Aligning data strategy with business goals
- Leading Data Governance Initiatives:
  - Establishing data governance frameworks
  - Promoting a data-driven culture
  - Managing data governance teams

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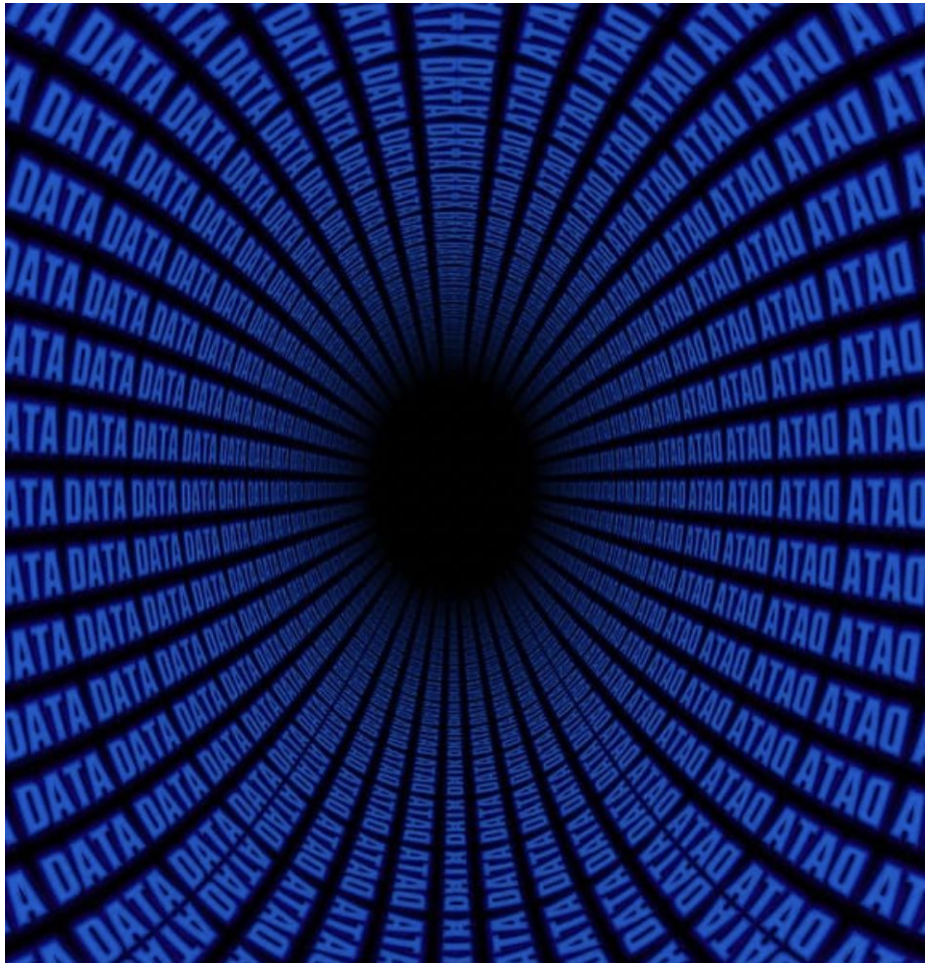
Module 4: Data Quality Management

Importance of Data Quality

- The impact of data quality on business operations
- Risks of poor data quality
- Benefits of high data quality



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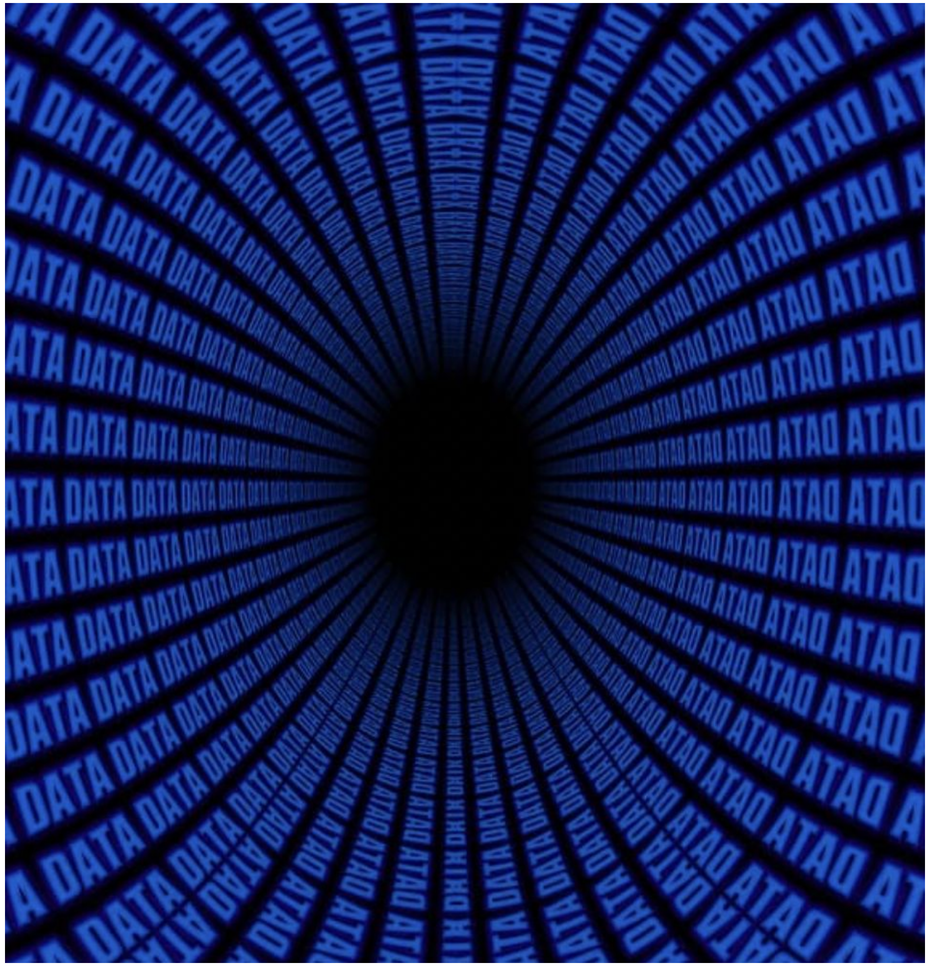
Module 4: Data Quality Management

Data Quality Dimensions

- Accuracy:
  - Ensuring correctness of data
- Completeness:
  - Addressing missing data
- Consistency:
  - Maintaining uniformity across datasets
- Timeliness:
  - Ensuring data is up-to-date and available



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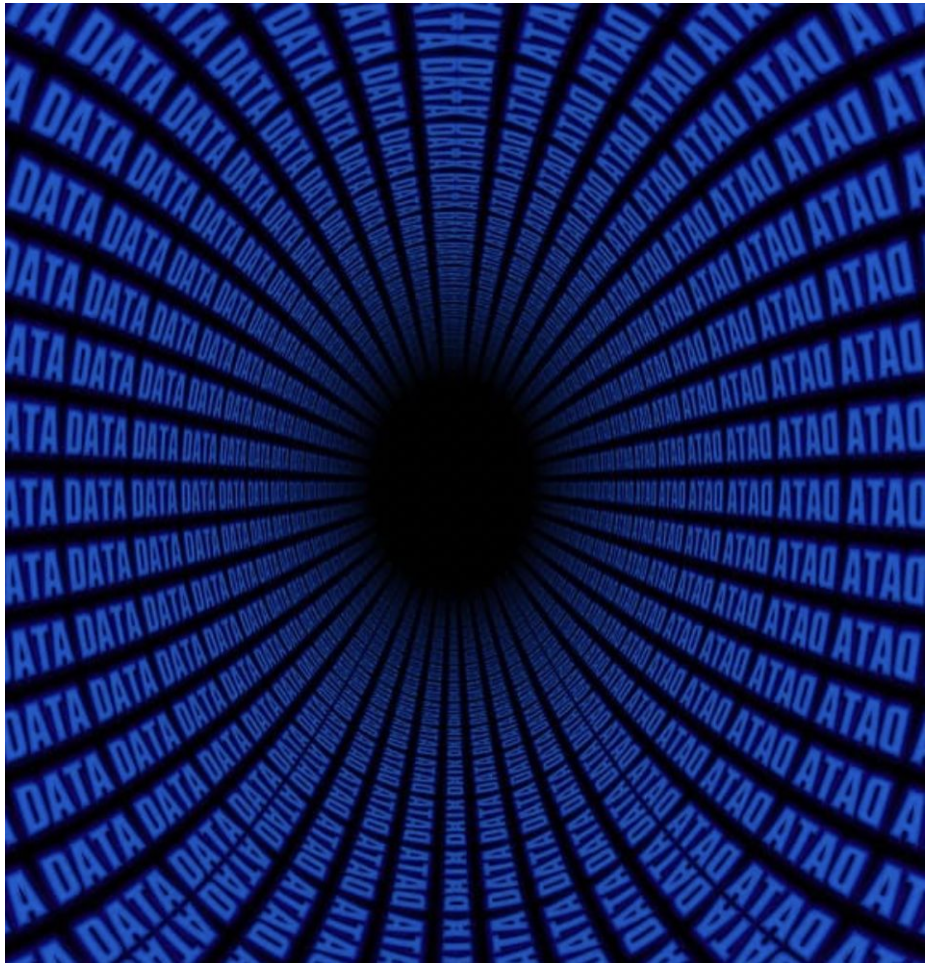
Module 4: Data Quality Management

Tools and Techniques for Data Quality Management

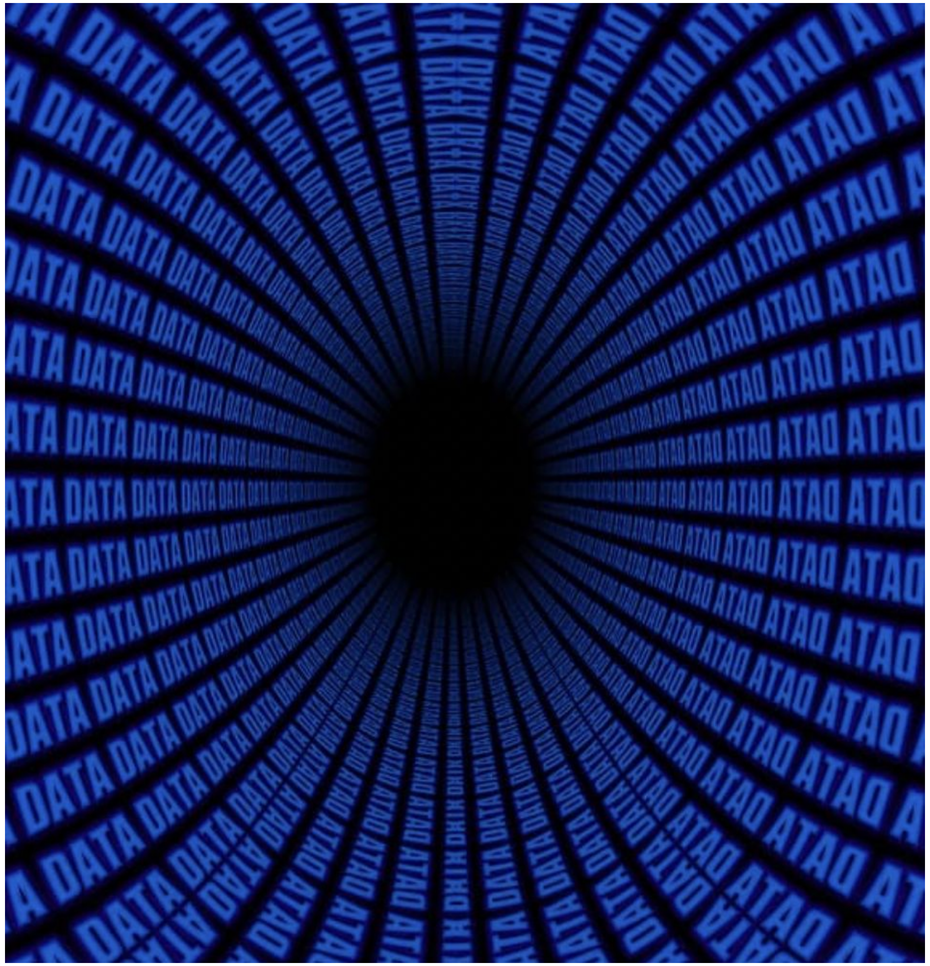
- Overview of Data Quality Tools (e.g., Talend, Informatica):
  - Features and capabilities
- Techniques for Assessing and Improving Data Quality:
  - Data profiling
  - Data cleansing
- Implementing Data Quality Solutions:
  - Steps and best practices



Module 4: Data Quality Management

Data Cleansing and Data Profiling

- Data Cleansing:
    - Identifying and correcting errors
    - Standardising data formats
- Data Profiling:
    - Analysing data for quality issues
    - Tools and methods for data profiling



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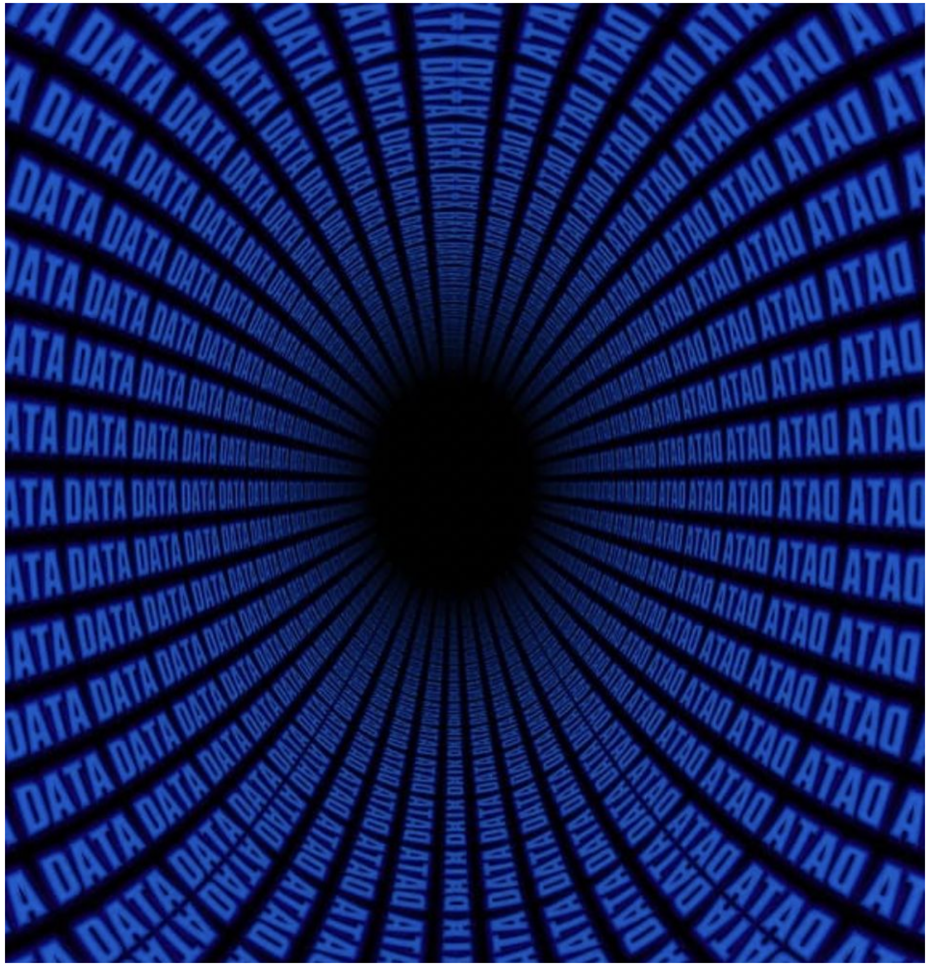
Module 4: Data Quality Management

Data Quality Rules and Implementation

- What are Data Quality Rules?
- Steps to Implement Your Own Data Quality Rules



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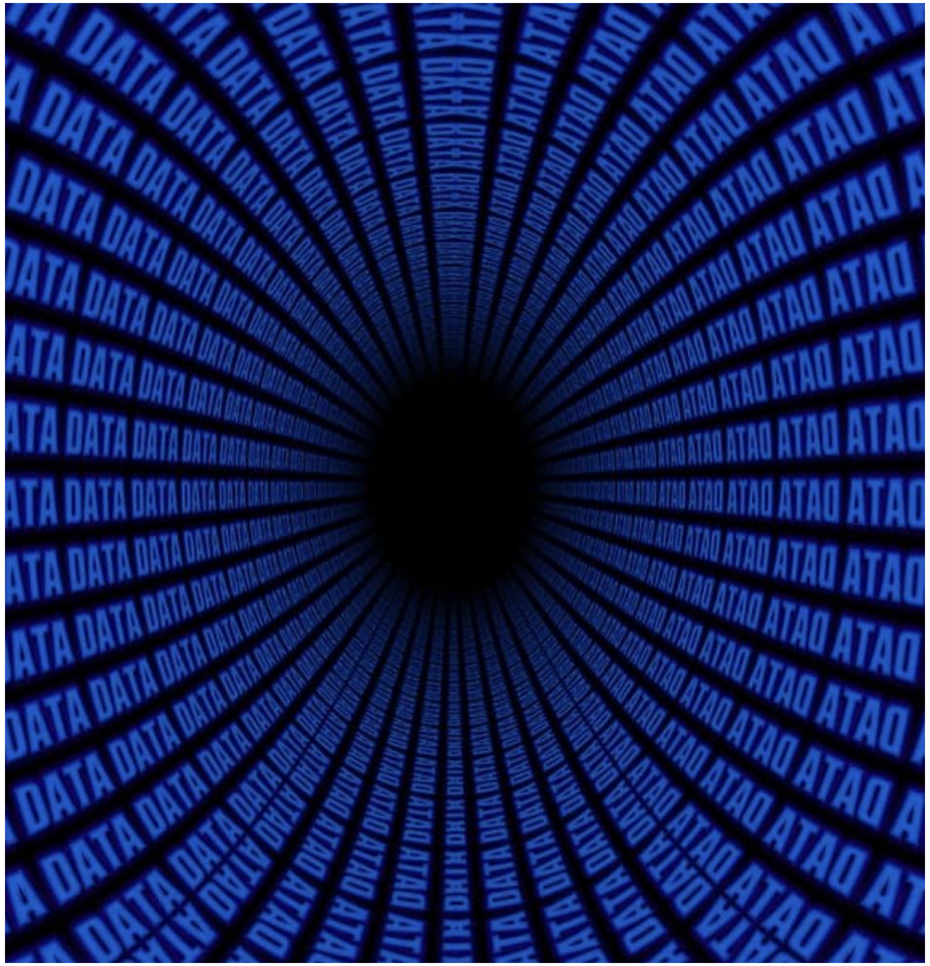
Module 4: Data Quality Management

Data Quality Management Process

- Define Data Quality Improvement Goals
- Data Profiling
- Root Cause Analysis Tools
- Resolve Data Quality Issues
- Monitor and Control



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Module 5: Data Management Fundamentals

Introduction to Data Management

- What is Data Management?
- Key Components of Data Management
- The Role of Data Management in Business Success

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Module 5: Data Management Fundamentals

Data Handling Ethics

- Ethical Principles in Data Handling
- Data Privacy and Security
- Compliance with Legal and Regulatory Requirements

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Module 5: Data Management Fundamentals

Data Architecture

- Fundamentals of Data Architecture
- Designing and Implementing Data Architecture

Module 5: Data Management Fundamentals

Data Modelling and Design

- Introduction to Data Modelling
- Designing Effective Data Models
- Tools and Techniques for Data Modelling

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Module 5: Data Management Fundamentals

Data Storage and Operations

- Data Storage Solutions
- Managing Data Operations
- Ensuring Data Availability and Reliability

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Module 5: Data Management Fundamentals

Data Security

- Key Principles of Data Security
- Implementing Data Security Measures
- Monitoring and Managing Data Security

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Module 5: Data Management Fundamentals

Data Integration and Interoperability

- Fundamentals of Data Integration
- Ensuring Interoperability of Data Systems

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Module 5: Data Management Fundamentals

Document and Content Management

- Importance of Document and Content Management
- Best Practices for Managing Documents and Content

Module 6: Data Governance and Management in Practice

Developing a Data Governance Plan

- Identify Data Governance Leadership & Scope
- Develop a Data Governance Charter
- Choose a Model for Data Governance
- Set Up the Data Governance Office
- Select the Data Governance Support Team

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Module 6: Data Governance and Management in Practice

Policy and Procedures Development

- Policy and Procedures for Data Governance
- Operational Data Governance Policies
- Data Compliance Policy
- Data Access Policy
- Data Security Policy
- Data Quality Policy
- Data Retention Policy
- Data Audit and Monitoring Policy
- Data Training and Awareness Policy

Module 6: Data Governance and Management in Practice

Measuring Data Governance Success

- What is a Data Governance Scorecard?
- Measuring the Data Governance Framework
- Measuring the Effectiveness of Data Stewardship
- Measuring Data Security and Privacy
- Measuring Data Catalog and Metadata Management
- Measuring Data Quality
- Measuring Communication and Training
- Measuring Data Governance Metrics and Reporting
- Measuring Data Issue Resolution
- Measuring Data Governance Tools and Technology
- Measuring Data Governance Audit and Compliance
- Measuring the Financial Impact

Module 7: Data Governance Tools and Technology

Overview of Data Governance Tools

- Essential for managing data quality, metadata, and compliance.
- Automate and streamline governance processes.
- Support roles such as data stewards and custodians.

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Module 7: Data Governance Tools and Technology

How to Select the Right Tool for Your Organisation

- Assess organisational needs and data governance objectives.
- Evaluate tool features: data lineage, metadata management, workflow automation.
- Ensure compatibility with existing systems.
- Analyse total cost of ownership and potential ROI.

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Module 7: Data Governance Tools and Technology

Popular Tools

- Collibra: Comprehensive data governance and catalog solution.
- Informatica: Strong in data integration and quality.
- Talend: Open-source data integration with governance features.
- IBM: Robust solutions with strong security features.
- Erwin by Quest: Data modelling and governance capabilities.
- ASG Technologies: Focuses on metadata management.
- SAP: Enterprise-grade data governance integrated with business applications.

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Module 7: Data Governance Tools and Technology

Steps for Successful Implementation

- Planning and Preparation
  - Define clear objectives and success criteria.
  - Assess current data governance maturity.
  - Develop a detailed implementation roadmap.

Module 7: Data Governance Tools and Technology

Steps for Successful Implementation

- Step-by-Step Implementation
  - Pilot Program: Start small to test and refine the tool.
  - Gather Feedback: Continuously collect input from users.
  - Full Rollout: Expand implementation organisation-wide.

Module 7: Data Governance Tools and Technology

Steps for Successful Implementation

- Training and Support
  - Provide comprehensive training for all users.
  - Ensure ongoing support and resources for troubleshooting.

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Module 7: Data Governance Tools and Technology

Integrating Tools with Existing Systems

- Compatibility and Integration
  - Assess compatibility with current IT infrastructure.
  - Plan for seamless integration with existing data systems.
- Data Migration
  - Ensure smooth data migration to new systems.
  - Minimise data loss and maintain data integrity.
- Interoperability
  - Enable interoperability between different data tools and systems.
  - Standardise data formats and protocols.

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Module 7: Data Governance Tools and Technology

Monitoring and Evaluating Tool Performance

- Performance Metrics
  - Define key performance indicators (KPIs) for tool effectiveness.
  - Regularly monitor and report on these metrics.
- User Feedback
  - Continuously gather and act on user feedback.
  - Use feedback to make iterative improvements.
- Continuous Improvement
  - Implement a process for ongoing evaluation and refinement.
  - Stay updated with new features and updates from tool providers.

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Congratulations on completing the Data Quality, Management, and Governance course!

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Next steps

- Apply what you've learned
- Leverage tools and technology
- Measure your success
- Review and reinforce your learning
- Further your learning
- Build your network
- Seek continuous improvement

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